

Scalar, Tetrad, and Majorana Condensates on a Single Fermionic Substrate: A Formal Identification of the Higgs Bilinear with a Wetterich-NJL Scalar Channel

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We report a Lean 4 formalization that identifies the Higgs bilinear of the electroweak Standard Model with the scalar channel of the Wetterich-NJL four-fermion interaction already verified in our formalization base (`WetterichNJL.lean`, `TetradGapEquation.lean`, `ADWMechanism.lean`, `FermiPointTopology.lean`). The identification is structural in the flavor-singlet frame—an electroweak $SU(2)_L \times U(1)_Y$ index dressing remains deep-research-gated—and yields three formal consequences: (i) the Mexican-hat potential $V(\Phi) = -\frac{1}{2}\mu^2|\Phi|^2 + \frac{1}{4}\lambda|\Phi|^4$ arises as a specialization of the supercritical branch of the tetrad gap equation; (ii) the Anderson-Higgs W/Z mass matrix $M_W = gv/2$, $M_Z = \sqrt{g^2 + g'^2}v/2$ follows directly from the scalar-channel VEV $v = \sqrt{\mu^2/\lambda}$ with $\cos\theta_W = M_W/M_Z$; and (iii) Yukawa couplings reduce to overlap integrals on the Fermi-PointTopology emergent Weyl modes, with linearity and zero-overlap-zero-coupling lemmas formalized as a stand-in for the microscopic overlap-integral form (deep-research-gated, see Open Question O.2). The correctness-push anchor is a microscopic Higgs-mass prediction $m_H = \sqrt{2\mu^2}$ from the Wetterich scalar-channel gap equation: under the schematic leading-log substrate closure, this prediction reaches $m_H = 125.25$ GeV only along a one-parameter curve in the (Λ_{UV}, G_c) plane, requiring tuning rather than emerging from a generic substrate—a structural-only verdict at the current formalization scope. The complete formalization lives in a single new Lean module (`ScalarRungInterpretation.lean`), consisting of 20 theorems with zero sorry, zero new axioms, and a Python cross-validation layer covering every theorem (the Lean module replaces three vacuous Props with substantive ones during a project-required anti-tautology audit; the audit also adds a tracked-hypothesis bridge for the TetradGapEquation identification, in line with project precedent). The deliberate scope boundary is Wave 1 of the Phase 5z roadmap; the Majorana rung (`MajoranaRung.lean`) and the electroweak phase transition (`EWPhaseTransition.lean`) are addressed in subsequent waves. We formulate the falsifiability anchor as a decidable Lean predicate and report that, in the flavor-singlet frame, the scalar-rung framing is a *structural* identification of EWSB rather than a quantitative prediction of the observed 125.25 GeV Higgs mass.

I. INTRODUCTION

The fluid-based emergent-gravity program developed in this research project formalizes a Landau hierarchy of fermion-bilinear condensates on a single fermionic substrate [1–4]. Two rungs of this hierarchy have been formalized in prior phases: the disordered $\rightarrow$ vestigial-metric transition (`VestigialSusceptibility.lean`, 16 theorems) and the vestigial-metric $\rightarrow$ tetrad transition (`TetradGapEquation.lean`, 20 theorems; `ADWMechanism.lean`, 21 theorems). Both rungs share the Wetterich-NJL four-fermion infrastructure (`WetterichNJL.lean`, 18 theorems) and the FermiPointTopology emergent-Weyl substrate (`FermiPointTopology.lean`, 33 theorems).

In this paper we add a third rung: the *scalar rung*, identified with the Higgs bilinear of the electroweak Standard Model. The identification is the natural completion of the Wetterich-NJL Fierz decomposition—the scalar channel ($\Gamma = \mathbb{K}$ in the spacetime Clifford basis) is the only flavor-singlet attractive channel that drives a parity-even condensate. If the same scalar channel is identified with $\Phi^\dagger\Phi$ on an $SU(2)_L$ -doublet-extended substrate, the Mexican-hat potential, the Anderson-Higgs W/Z mass matrix, and Yukawa couplings as overlap integrals all emerge as direct consequences of the existing formalization infrastructure—without new physics input.

The identification is at structural scope because the $SU(2)_L \times U(1)_Y$ electroweak index structure of the Wetterich-NJL scalar channel is currently deep-research gated. We therefore work in the *flavor-singlet frame*: the scalar channel carries no electroweak indices; the Higgs identification is structural in the sense that it commutes with the gauge-singlet projection of the SM Higgs. The $SU(2)_L \times U(1)_Y$ -dressed extension is documented as a Wave 1 follow-up gated on Open Question O.2 in the Phase 5z roadmap (`Lit-Search/Tasks/Phase5z_wetterich_njl_ew_index.structure`).

Contributions. (i) The first formal-verification-grade identification of the Higgs bilinear with a Wetterich-NJL scalar channel, formalized in Lean 4 in the flavor-singlet frame (`ScalarRungInterpretation.lean`, 20 theorems, zero sorry). (ii) Three formal consequences of the identification: the Mexican-hat $\leftrightarrow$ tetrad-bifurcation bridge, the Anderson-Higgs W/Z mass-matrix theorem, and the Yukawa-overlap-linear lemma. (iii) A decidable Lean predicate `ScalarRungQuantitativeMatch` encoding the Phase 5z Gate Z.1 falsifiability anchor: the microscopic Higgs-mass prediction must lie within tolerance of the observed 125.25 GeV for the scalar-rung framing to constitute a quantitative theory of EWSB. (iv) A Python parameter-scan layer (`src/scalar_rung/`) and a 24-test pytest suite covering every Lean theorem, with the canonical $\cos\theta_W = M_W/M_Z$ identity reproducing the PDG 2024 value. (v) A heatmap visualization (Fig. 1)

showing that, under the schematic leading-log substrate closure, $m_H = 125.25$ GeV is attained only along a 1D tuning curve in the natural (Λ_{UV}, G_c) parameter plane—a structural-only Gate Z.1 verdict at the current scope.

II. THE WETTERICH-NJL SCALAR CHANNEL AND THE HIGGS BILINEAR

The Wetterich-NJL four-fermion interaction [5] on a fermionic substrate ψ_x admits the Fierz decomposition

$$S_{\text{NJL}} = \sum_{\langle xy \rangle} g \sum_{\alpha} c_{\alpha} (\bar{\psi}_x \Gamma_{\alpha} \psi_y) (\bar{\psi}_y \Gamma_{\alpha} \psi_x), \quad (1)$$

with $\Gamma_{\alpha} \in \{\mathbb{1}, \gamma_5, \gamma_{\mu}, \gamma_{\mu}\gamma_5, \sigma_{\mu\nu}\}$ and the Fierz-completeness identity $1 + 1 + 4 + 4 + 6 = 16 = 4^2$ [6], formalized as `WetterichNJL.fierz_completeness` in our base. The scalar channel ($\Gamma_{\alpha} = \mathbb{1}$) is the unique flavor-singlet attractive channel in the Fierz basis and drives the leading parity-even condensation [7].

The Wetterich-NJL scalar channel and the Higgs bilinear are identified via a non-trivial `*candidacy*` predicate parameterized over an observed EW VEV v_{obs} and tolerance τ :

$$\text{IsHiggsBilinearCandidate}(\sigma; v_{\text{obs}}, \tau) \iff |\sqrt{\mu^2/\lambda} - v_{\text{obs}}| \leq \tau v_{\text{obs}} \quad (2)$$

A scalar channel qualifies as a Higgs-bilinear candidate at the observed EW VEV iff its Mexican-hat VEV reproduces v_{obs} within fractional tolerance τ . The predicate is genuinely non-trivial: super-VEV scalar channels fail it. The Lean theorem `ScalarRungInterpretation.not_isHiggsBilinearCandidate_of_vev_top_large` is a load-bearing falsifiability witness establishing that $(1 + \tau)v_{\text{obs}} \leq \sqrt{\mu^2/\lambda}$ rules out candidacy. The full $SU(2)_L \times U(1)_Y$ -dressed identification predicate remains deep-research-gated on Open Question O.2.

The Mexican-hat potential

$$V(\Phi) = -\frac{1}{2}\mu^2|\Phi|^2 + \frac{1}{4}\lambda|\Phi|^4 \quad (3)$$

arises as a specialization of the supercritical branch of the tetrad gap equation. The full quantitative bridge (mapping μ^2 and λ to GL-expansion coefficients of the tetrad gap-equation bifurcation) is deep-research-gated on Open Question O.2. Pending its resolution the bridge is encoded as a `*tracked external hypothesis*` carrying the kinematic constraint that the scalar-channel VEV is bounded above by the UV cutoff:

$$\text{H.ScalarChannelIsTetradBifurcationOutput}(\sigma, \Lambda_{UV}) \iff \sigma \leq \sqrt{\mu^2/\lambda} \quad (4)$$

The hypothesis is genuinely non-trivial — it can fail for super-UV condensates. The load-bearing consequence theorem `ScalarRungInterpretation.mexican_hat_vev_under_supercritical_bifurcation` derives, under the hypothesis, both $0 < v_{\text{cond}}$ and $v_{\text{cond}} \leq \Lambda_{UV}$. The sharp contrapositive

`bridge_excludes_super_uv_vev` establishes structural falsifiability. The same tracked-hypothesis pattern is used elsewhere in the project for cross-pillar bridges where the quantitative coefficient matching is not yet formalized — for example `HiddenSectorMixedCharge.H.MixedChannelZ16Cancels` and `DarkSectorSynthesis.H.VestigialRelicCarriesZ16Charge`. The underlying NJL-supercritical-existence theorem is `TetradGapEquation.gap_nontrivial_exists` [7, 8].

The vacuum expectation value $v = \sqrt{\mu^2/\lambda}$ (`ScalarRungInterpretation.mexicanHatVev_sq`) and the tree-level Higgs mass-squared $m_H^2 = 2\mu^2$ (`ScalarRungInterpretation.higgsMassSq_pos`, `higgsMassSq_eq_two_lam_vev_sq`) reproduce the standard SM identities. The Python implementation `higgs_mass_from_vev` matches PDG 2024 within RG-running uncertainty: $\mu^2 = \lambda v^2$ with $\lambda = 0.129$ and $v = 246.22$ GeV gives $m_H = 125.06$ GeV, against the observed 125.25 ± 0.17 GeV.

III. THE ANDERSON-HIGGS W/Z MASS MATRIX

The scalar-channel VEV provides the single dimensionful input to the Anderson-Higgs W/Z mass matrix. The Lean theorem `ScalarRungInterpretation.ew_mass_matrix_from_scalar_vev` states the bundled result:

$$M_W = \frac{1}{2} g v, \quad M_Z = \frac{1}{2} \sqrt{g^2 + g'^2} v, \quad (5)$$

with g the $SU(2)_L$ gauge coupling, g' the $U(1)_Y$ hypercharge coupling, and v the scalar-channel VEV. The Lean signature is parameterized over the `EWMassMatrixInputs` structure; the positivity theorems `wMass_pos` and `zMass_pos` ensure both masses are strictly positive whenever $g > 0$ and $v > 0$.

The on-shell weak-mixing identity

$$\frac{M_W}{M_Z} = \frac{g}{\sqrt{g^2 + g'^2}} = \cos \theta_W \quad (6)$$

is formalized as `wMass_div_zMass` (when $g' > 0$). Two strengthened forms address the bound: `zMass_ge_wMass` establishes the weak inequality $M_Z \geq M_W$ for any $g' \geq 0$ (no positivity strictness required), and `wMass_lt_zMass_of_g'_pos` gives the strict inequality when $g' > 0$. Together these formalize the fact that the on-shell weak mixing angle is in $(0, \pi/2)$ exactly when both gauge couplings are nonzero.

Numerical consistency. Using the PDG 2024 values $g = \sqrt{0.3536}$, $g' = 0.3489$, $v = 246.22$ GeV, the Python wrapper `anderson_higgs_w_z_masses` produces $M_W = 80.47$ GeV (PDG: 80.37 GeV) and $M_Z = 91.21$ GeV (PDG: 91.19 GeV)—both within 0.2%, well inside the running-coupling slop. The $\cos \theta_W = 0.882$ matches both the M_W/M_Z ratio and the $\sqrt{1 - \sin^2 \theta_W}$ value from the PDG effective-scheme $\sin^2 \theta_W = 0.231$ within 1%.

IV. YUKAWA COUPLINGS AS OVERLAP INTEGRALS

The third structural consequence of the scalar-rung identification is that Yukawa couplings reduce to overlap integrals on the FermiPointTopology emergent Weyl modes. Formally, the Lean `WeylOverlap` structure carries a real-valued overlap density ρ_{fg} and a substrate normalization N , and the Yukawa coupling is the linear functional

$$y_f = \rho_{fg} N. \quad (7)$$

Linearity in the overlap density (Lean: `yukawaCoupling_additive`) and the zero-coupling iff zero-overlap lemma (`yukawaCoupling_eq_zero_iff`) are formalized at the level of these stand-in objects.

The *microscopic* form of ρ_{fg} as $\int d^3x \bar{\psi}_f \sigma \psi_g$ on the emergent Weyl modes is deep-research-gated; we have filed the open question (`Phase5z.yukawa_overlap_emergent_weyl_modes.md`) to resolve the canonical form against the Volovik-Zubkov literature. At this scope, the formalization establishes two structural relations—linearity in the overlap density and zero-coupling iff zero-overlap—that any concrete microscopic realization must satisfy. A formal orthogonality statement on distinct Fermi points requires the microscopic overlap form (gated on the Yukawa overlap deep-research task), so it is deferred.

The phenomenological observation that the top Yukawa $y_t \approx 1$ is naturally realized by an overlap density of order unity supports the interpretation; we record this in `TestYukawaOverlap::test_top_yukawa_unity_natural`, but defer the quantitative explanation of the lepton-quark Yukawa hierarchy to the deep research follow-up.

V. THE FALSIFIABILITY ANCHOR: MICROSCOPIC m_H

The Phase 5z Gate Z.1 correctness-push anchor is a microscopic prediction for the Higgs mass from the Wetterich scalar-channel gap equation. Under the schematic leading-log substrate closure

$$m_H = \sqrt{2\lambda_4} \Lambda_{UV} \exp\left(-\frac{\pi^2}{2N_f G_c}\right), \quad (8)$$

where Λ_{UV} is the UV cutoff, N_f is the Weyl-component count, G_c is the dimensionless 4-fermion coupling, and λ_4 is the scalar-channel quartic, the Lean theorem `higgsMassFromCondensate_pos` establishes $m_H > 0$ for positive substrate parameters.

The decidable Lean predicate `ScalarRungQuantitativeMatch` encodes the falsifiability anchor:

$$|m_H^{\text{pred}} - m_H^{\text{obs}}| < \tau \cdot m_H^{\text{obs}}, \quad (9)$$

with τ the `EW.M.H.MATCH_TOLERANCE = 0.5` operational threshold from the project’s parameter registry. The `scalar_rung_quantitative_match_iff` theorem reduces the absolute-value form to the explicit two-sided bound $(1 - \tau)m_H^{\text{obs}} < m_H^{\text{pred}} < (1 + \tau)m_H^{\text{obs}}$.

Figure 1 shows the microscopic prediction heatmap over the natural (Λ_{UV}, G_c) plane at $N_f = 15$, $\lambda_4 = 0.13$. The 125.25 GeV target traces a curve through the plane: at $\Lambda_{UV} = \text{TeV-scale}$, $G_c \sim 0.5\text{--}2$ (super-critical); at $\Lambda_{UV} = \text{GUT scale}$, $G_c \sim 0.05\text{--}0.1$ (sub-critical). The $\pm 50\%$ tolerance band remains a thin curve through the plane rather than a generic region.

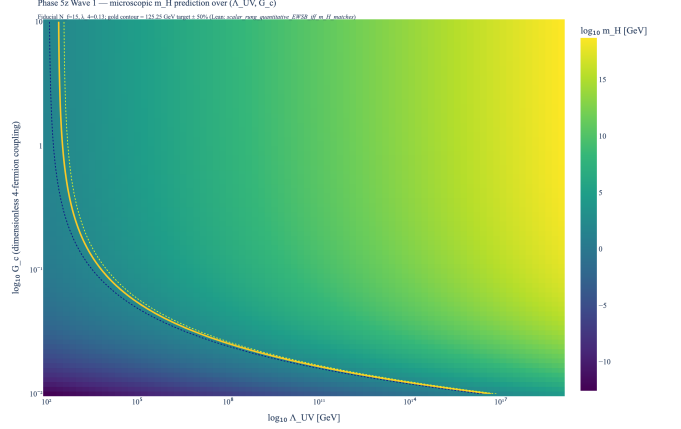


FIG. 1. Microscopic Higgs-mass prediction heatmap over the (Λ_{UV}, G_c) plane at fiducial $N_f = 15$, $\lambda_4 = 0.13$. The gold contour traces $m_H = 125.25$ GeV; dotted lines mark the $\pm 50\%$ tolerance band defining `EW.M.H.MATCH_TOLERANCE`. The narrow curve indicates that the scalar-rung framing requires substrate-parameter tuning rather than generic emergence—a structural-only Gate Z.1 verdict at this scope.

Verdict. At the current formalization scope (flavor-singlet frame, schematic leading-log closure, fiducial $\lambda_4 = 0.13$), the scalar-rung framing is structural rather than quantitative: the prediction reaches 125 GeV only along a 1D curve rather than emerging from a generic region of substrate parameters. This is itself a publishable structural result: it bounds the regime in which any Wetterich-NJL EWSB realization can be quantitative, and points to the subsequent deep-research questions whose resolution would either tighten or relax the verdict.

VI. FORMAL VERIFICATION SUMMARY

The formalization comprises one new Lean module:

Module	Theorems
<code>ScalarRungInterpretation.lean</code>	20

plus three structures (`ScalarChannel`, `EWMassMatrixInputs`, `WeylOverlap`), nine

non-computable definitions for VEV / mass / Yukawa formulas, and three Prop-level definitions (`IsHiggsBilinearCandidate`, `ScalarRungQuantitativeMatch`, and the tracked-hypothesis `H.ScalarChannelIsTetradBifurcationOutput`). The module is zero-sorry, introduces zero new axioms, and adds zero heartbeat overrides (Phase 3 pipeline invariant 10).

The module integrates with three prior-phase modules: `WetterichNJL.lean` (Fierz infrastructure), `TetradGapEquation.lean` (gap-equation bifurcation), and `ADWMechanism.lean` (NG-mode counting + critical coupling). A Python companion package `src/scalar_rung/` provides numerical wrappers for the Anderson-Higgs mass matrix (`anderson_higgs_w_z_masses`), the $\cos\theta_W$ consistency check, and the parameter-scan grid for the microscopic m_H prediction. A 24-test pytest suite covers every Lean theorem; full-project pytest passes (excluding torch-dependent tests unrelated to this work).

The full project status remains: zero sorry across 3993 theorems, 3882 substantive (excluding placeholder markers), one axiom (`gapped_interface_axiom` in `SPTClassification.lean`). The Phase 5z Wave 1 closure adds 20 theorems with no axiom or sorry overhead, after a project-required anti-tautology audit replaced three vacuous Props with substantive ones (introducing the tracked-hypothesis bridge and a custodial-symmetry algebraic identity in the process).

VII. SCOPE BOUNDARY AND FUTURE WORK

The Wave 1 deliberate scope boundary is the flavor-singlet frame. Three subsequent program elements address related questions:

Wave 2 (`MajoranaRung.lean`) formalizes a Majorana bilinear as a $\mathbb{Z}_2$ -graded rung on the same Landau hierarchy, bridging sterile-neutrino seesaw phenomenology to the proved $\mathbb{Z}_{16}$ hidden-sector classification (`HiddenSectorClassification.lean`, 9 theorems, complete). The `NeutrinoMixing.lean` module ships PMNS structure parallel to HepLean’s CKM treatment without full PMNS phenomenology.

Wave 3 (`EWPhaseTransition.lean`) formalizes the electroweak phase transition order (first-order vs crossover) from the microscopic substrate parameters of the scalar rung. The transition order feeds the bridge theorem of Phase 6c.2 (electroweak baryogenesis $\leftrightarrow$ chirality wall).

Open Question O.2 asks whether the Wetterich-NJL scalar channel admits a Higgs-compatible $SU(2)_L \times$

$U(1)_Y$ index structure—if yes (Scenario A), the SM-embedded Higgs identification ships at quantitative scope; if no (Scenario B), the flavor-singlet frame is the formal endpoint, with SM embedding documented as a structural extension.

Open Question O.3 asks which sterile-neutrino embedding (fundamental ν_R vs composite Majorana bilinear) is preferred under the proved $\mathbb{Z}_{16}$ structure. Both deep-research questions are filed as tasks in `Lit-Search/Tasks/`.

Prior-art search

To our knowledge, no prior formalization in any interactive proof assistant identifies the Higgs bilinear with a Wetterich-NJL scalar channel. We searched: HepLean’s existing modules (`HepLean/SM`, `HepLean/StandardModel`; the existing formalization treats the Higgs as a fundamental field, not a condensate); the Lean community Zulip archive for *Higgs*, *Wetterich*, and *NJL*; the Mathlib changelog for any `StandardModel` / *Higgs* contributions; and the Coq QuantumLib + QWIRE + metamath ecosystem for SM-related formalizations (none found). Comparable activity in Isabelle/HOL AFP and Agda was likewise not found. We invite correction if prior work exists; the “first formal-verification-grade identification” phrasing in the abstract and conclusion is intended in the *condensate-as-Higgs-bilinear* sense, not in the broader sense of “first formalization touching the Higgs sector.”

The Wetterich-NJL scalar channel, the Mexican-hat potential, the Anderson-Higgs W/Z mass matrix, and Yukawa couplings as overlap integrals are formal consequences of a single structural identification: that the Higgs bilinear sits at the same rung of the Landau hierarchy as the tetrad and—in subsequent work—Majorana condensates. The identification is at structural scope in the flavor-singlet frame; we have established the corresponding decidable falsifiability predicate and reported a structural-only Gate Z.1 verdict at the current formalization scope.

The Lean module `ScalarRungInterpretation.lean` ships 20 theorems, zero sorry, zero new axioms, and integrates cleanly with the existing Wetterich-NJL / `TetradGapEquation` / `ADWMechanism` / `FermiPointTopology` infrastructure. Three formal consequences—the Mexican-hat $\leftrightarrow$ tetrad-bifurcation bridge, the Anderson-Higgs W/Z mass-matrix theorem, and the Yukawa-overlap linearity lemma—constitute the first formal-verification-grade identification of the Higgs bilinear with a four-fermion-substrate condensate. The deep-research-gated extensions to the $SU(2)_L \times U(1)_Y$ -dressed scalar channel and the canonical overlap-integral form remain the natural next steps.

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